

Quick Activity 1: School Supplies Calculator

1. Problem Analysis

The program calculates the total cost of all school supplies by multiplying the quantity of each item by its price, then adding all the subtotals together.

The program should determine:

- The cost of each school supply.
- The total amount to be paid.

2. Inputs-Process-Outputs

Input	Process	Output
Quantity and prices of Pens	Multiply quantity x price	Pen subtotal
Quantity and prices of Notebook	Multiply quantity x price	Notebook subtotal
Quantity and prices of Eraser	Multiply quantity x price	Eraser subtotal
Quantity and prices of Pencils	Multiply quantity x price	Pencils subtotal
All subtotal	All subtotal	All subtotal

Formula:

Subtotal = Quantity = Unit Price

Total Cost = Sum of all Subtotals

3. Decomposition

The problem can be divided into these smaller tasks.

1. Get the quantity and unit price of each school supply
2. Calculate the subtotal for each item.
3. Add all the subtotal.
4. Display the total cost.

4. Pseudocode

START

INPUT pen_quantity

INPUT pen_price

INPUT notebook_quantity

INPUT notebook_price

INPUT eraser_quantity

INPUT eraser_price

INPUT pencils_quantity

INPUT pencils_price

pen_total = pen_quantity x pen_price

notebook_total = notebook_quantity x notebook price

eraser_total = eraser_quantity x eraser_price

pencils_total = pencils_quantity x pencils_price

Total_cost = pen_total + notebook_total + eraser_total + pencils_total

DISPLAY pen_total

DISPLAY notebook_total

DISPLAY pencil_total

DISPLAY eraser_total

DISPLAY total_cost

END

5. Dry Run - Test Case 1

Given:

5 pens x ₱15 = ₱75

3 notebooks x ₱ 20 = ₱60

3 pencils x ₱10 = ₱30

2 eraser x ₱6 = ₱12

Calculation:

item	Quantity	Unit Price	Subtotal
Pen	5	₱15	₱120
Notebook	3	₱20	₱60
Pencils	3	₱10	₱30
Eraser	2	₱6	₱12

Total Cost = ₱120 + ₱60 + ₱30 + ₱12

Total Cost = ₱322

Output:

Total cost: ₱322

6. Dry Run - Test Case 1

Given:

- 4 pens x ₱15 = ₱60
- 5 notebooks x ₱ 20 = ₱100
- 6 pencils x ₱10 = ₱60
- 3 eraser x ₱6 = ₱18

Calculation:

item	Quantity	Unit Price	Subtotal
Pen	4	₱15	₱60
Notebook	5	₱20	₱100
Pencils	6	₱10	₱60
Eraser	3	₱6	₱18

Total Cost = ₱60 + ₱100 + ₱60 + ₱18

Total Cost = ₱238

Output:

Total cost: ₱238